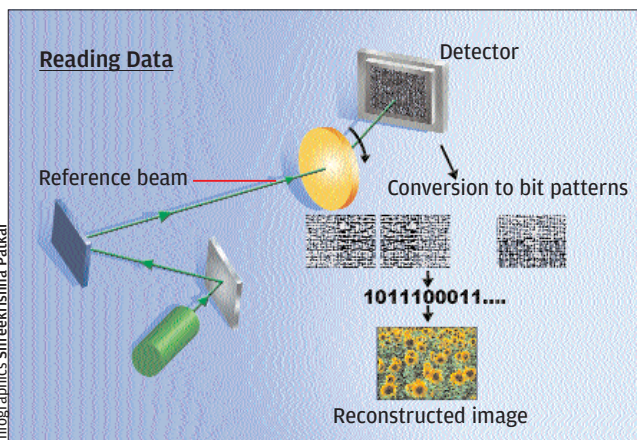
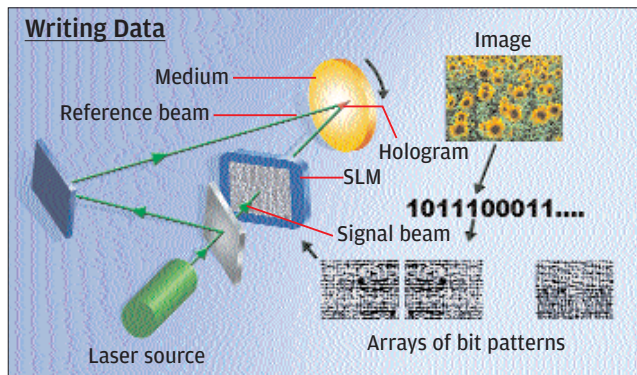


Holographic Storage

Holographic Storage has been talked about for a long time; indeed, the Holographic Versatile Disc (HVD) is being eagerly awaited by people around the world. This technology uses lasers to record data in the *volume* of the medium, rather than on the surface. The idea, surprisingly, is not new, but it's only now that the technology seems to be getting up to speed.

How Does It Work?

A laser beam is split in two, the reference beam and the signal beam (called so because it carries the data). A device called a spatial light modulator (SLM) translates 1s and 0s into



an optical pattern of light and dark pixels, as in the figure above. These pixels are arranged in an array (or "page") of about a million bits.

The signal beam and the reference beam intersect in the storage medium, which is light-sensitive. And at the point of intersection, a hologram is formed because of a chemical reaction in the medium, and gets recorded there. (A hologram is the interference pattern that results when two light waves meet; for more on interference, visit www.physicsclassroom.com). For reading the data, only the reference beam is used: it deflecks off the hologram, and a detector picks up the data pages in parallel. The 1s and 0s of the original data can be read from the data pages.

By varying the angle or wavelength of the reference beam, or by slightly changing the placement of the media, lots of holograms can be stored in the volume of the medium.

What Work Is On?

Optware Corporation of Japan have already come out with their HVD: the HVD holds 1 TB (a terabyte), and is the same size as a regular optical disc. Enterprise versions were planned for this year: the estimated costs were something

like \$20,000 (Rs 9 lakh) for players and \$100 (Rs 4,500) for discs.

In the meanwhile, InPhase Technologies, Optware's main competitor, is coming out with products of its own. In partnership with Maxell, InPhase has already come up with a 300 GB disk, with an 800 GB disc expected in 2008. And if the 1 TB of HVD weren't enough, Fuji Photo Film USA has demonstrated a type of HVD with a claimed capacity of 3.9 TB!

What Promise Does It Hold?

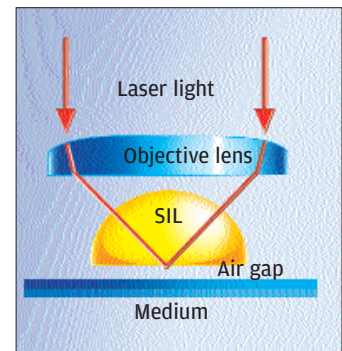
Besides high storage densities, holographic storage means fast access times, because there are no actuators as in hard disks—laser beams can be focused around much more rapidly.

Holographic storage can kill off the hard disk within the next 10 years, but that's again speculation—some technologies step out quickly and graciously, but some are pretty stubborn!

Near-Field Optical Recording Using A Solid Immersion Lens

Think of a CD or DVD; while recording, the lens focuses the laser onto a tiny spot on the medium. This spot is tinier for DVD than for CD, and is even tinier in Blu-ray, for example.

Near-field optical recording (NFOR) refers to the extremely sharp focusing of a laser beam, which means an extremely small distance between the lens and the recording medium. NFOR using a solid immersion lens (SIL) would be the child of Blu-ray and HD-DVD, and therefore, the grandchild of the DVD.



How Does It Work?

The density of the data that can be achieved on a disc is roughly proportional to the square of the numerical aperture (NA) of the lens, and inversely proportional to the wavelength of the laser (refer *The Battle Of The Blue*, Digit December 2005). The NA of a lens dictates how sharply it can focus the beam falling on it. The NA of a SIL is made very high, and the achievable data densities are therefore that much higher.

In NFOR using a SIL (refer figure alongside), the laser is very sharply focused: it converges at a point within the lens, instead of on the medium. The air gap between the lens and the medium is just about 25 nm! The photons "tunnel" through the air gap onto the surface of the medium (See *Jargon Buster*).

What's Being Done?

About half a year ago, Philips researchers reported "significant progress" in developing NFOR. Up to 150 GB of data on a dual-layer disc would be possible, they said, although they also said the technology was several years away from commercialisation.

What Promise Does It Hold?

NFOR using SILs seems in no way to us a hard disk killer, but it's the natural progression from blue-laser systems. NFOR is therefore a potential "Blu-ray killer." The promise NFOR holds all seems to hinge on how much comes in in terms of research dollars. Will the HVD have already taken over by the time NFOR takes off?